

Chapter – 3: Landforms and Life

Free from the burden of human beings, may the Earth with many heights, slopes and great plains, bearing plants endowed with varied powers, spread out for us and show us her riches! ... The Earth is my mother and I am her child. — Atharva Veda, Bhūmi Sūkta ('Hymn to the Earth')

Introduction

- Humans – along with – most mammals – live on land
- Land – many forms, features – appearance change – region to region
- Travel – Chhota Nagpur in Jharkhand to Prayagraj in Uttar Pradesh to Almora in Uttarakhand
- On the way – very diff. landscapes – to be specific – 3 major landforms
- Landform – phy. feature – surface – planet Earth
- Landforms – take millions of years – significant connection – env. and life
- Divided into 3 categories – **mountains, plateaus and plains**
- These landforms – diff. climate – variety – flora and fauna
- Humans – adapted – all landforms – BUT – no. of people – on diff. landforms – varies too much

Mountains

- Landforms – much higher – surrounding landscapes
- Recognise – broad base – steep slope – narrow summit
- Depending on height – some mountains – covered with snow
- Lower **altitudes** – snow melts – every summer – feeds rivers
- High alt. – snow may never melt – mountains – snow-capped always
- Other highlands – lower height – less steep slope – rounded tops – **hills**
- Snow – same as rainfall – solid state
- Most mountains – grouped – **mountain ranges** –
 - Asia – Himalayas
 - Europe – Alps
 - South America – Andes



- Some ranges – stretch – thousands of kms
- 6 mountains – visual impression – rel. heights
 - Mt. Everest {b/w Tibet (China) and Nepal} and Kanchenjunga (b/w Nepal and Sikkim) – 2 highest peaks – Himalayan range
 - Mt. Aconcagua (South America) – highest – Andes
 - Mt. Kilimanjaro – eastern Africa – isolated – no range
 - Mt. Blanc (Western Europe) – highest – Alps
 - Anamudi (Kerala) – Anai Peak – highest – South India
- Mountains – tall, sharp peaks – Himalayas – rel. 'young' – formed much recently in Earth's history – some millions of years ago
- Shorter, rounded – mountains, hills – Aravalli Range – much older – rounded by erosion
- Sometimes – like Himalayas – upliftment, erosion – still continues



Mountain Environment

- Mountain slopes – covered – forest – **montane forest** – conifer trees – pines, firs, spruce, deodar – common
- Conifer trees – grow tall – cone-shaped – thin, pointed leaves
- Higher alt. – trees give way – grasses, **mosses** and **lichens**
- 2 verses – long poem by *Kalidasa* – 1500 yrs. ago
- Poem – *Kumarasambhava* – begins – calling to Himalayas –
 - In the north rises Himālaya, the Lord of mountains, like a living god, who measures the Earth and stretches from the western to the eastern oceans. ...
 - From it the wind comes down, carrying spray from descending Gangā, shaking the deodar trees, opening the peacocks' tail feathers and cooling the mountain people after they hunt deer.
- Deep forests – flowing rivers – lakes – grasslands – caves – diverse fauna – golden eagle, peregrine falcon, Canadian lynx, snow leopard, ibex, Himalayan tahr, mountain hare, yak, grey fox, black bear
- Many rivers – originate – mountains –
 - Ganga – largest river – Himalayas – 2500 km – numerous tributaries
 - Yamuna, Ghagara – Himalayas
 - Son (Sone) – Vindhya Range

Life in the Mountains

- Mountain **terrain** – rugged (rough) – steep slopes
- Regular farming – practiced – some **valleys**
- Cultivation – slopes – cutting steps – terrace farming
- Many mountain regions – herding preferred over agriculture
- Tourism – imp. src. – income – mountains
- Crisp mountain air – scenic beauty – attract many tourists
- Some tourists – sports – skiing, hiking, mountaineering, paragliding
- Many centuries – people – travel – pilgrimage – holy site
- Excessive tourists – mountain env. – under stress – difficult to find balance

Plateaus

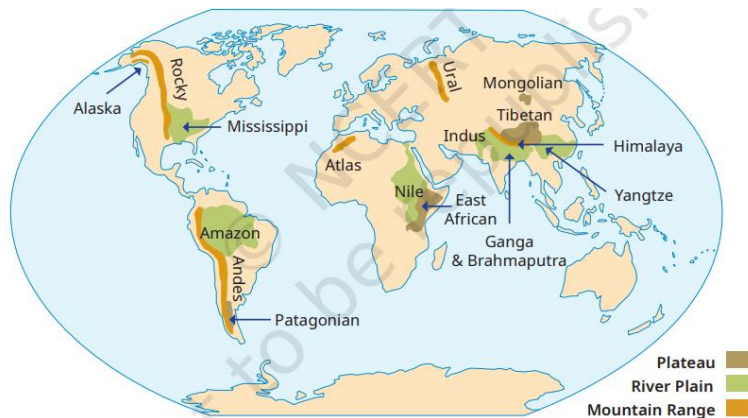
- Landform – rises up – surrounding land – more or less flat surface – sides – steep slopes
- Plateaus – also young or old – Earth's history
- 2 examples –
 - Tibetan Plateau – largest and highest plateau
 - Avg. alt. – 4500 m – nickname – Roof of the World
 - East to west – 2500 km
 - Deccan Plateau
 - One of the oldest
 - Formed – volcanic activity millions of years ago
- Height – vary – few hundred to few thousand meters
- Plateaus – rich in mineral deposits – storehouse of minerals
- Mining – major activity – many – world's largest mines
- Example –
 - East African Plateau – gold and diamond mining
 - Chhota Nagpur Plateau – iron, coal and manganese
- Env. – very diverse – most plateaus – rocky soil – less fertile than plains – less favourable for farming

- Exception – lava plateau – result of volcanic activity – often – rich black soil
- Plateaus – spectacular waterfalls
 - Victoria falls – Zambezi River – southern Africa
 - Hundru falls – Subarnarekha River – Chhota Nagpur Plateau
 - Jog falls – Sharavati River – Western Ghats
 - Nohkalikai falls – 340 meters – Cherrapunji Plateau (Meghalaya)



Plains

- Landforms – extensive flat surface – no large hills – no deep valleys
- Generally – under 300 meters above **sea level**
- Floodplains – formed by – rivers – originating from mountains – collect – particles – rock, sand, silt – sediments
- Sediments – carried to plains – deposited by rivers – soil becomes very fertile
- Result – these plains – ideal – farming – all kinds
- Agriculture – major occupation
- Also support – variety – flora and fauna



Life in the plains

- 1000s of yrs. ago – 1st civilisations – settled in fertile plains – around rivers
- Our times – large part – world's population – fertile plains
- 40 cr. – more than 1/4th of total Indian population – India's Ganga Plain (Gangetic plain)
- Major occupation – river fishing and agriculture
- Food crops – rice, wheat, maize, barley and millets
- Fibre crops – cotton, jute and hemp
- Traditional agriculture – rainfed – watered by rainfall
- Recent decades – agriculture – turned to irrigation – water comes to field – networks of canals or groundwater
- Irrigation – increased – agriculture – BUT – decreased – groundwater – challenge for future agriculture
- Problems – high population and pollution
- Rivers – carry – immense cultural value
- Many communities – river's origin and **confluence** – sacred
- India – numerous festivals, ceremonies, rituals – conducted – such locations
- Plains – gentle slopes – river navigation – easier – supports – economic activities
- Earlier days – people – used rivers – travel



- Even today – stretches – along Ganga – people prefer – boats – move around
- This chapter – 3 main landforms
- BUT – Earth's surface – very complex – experts – define – few more landforms
- Example – desert – large and dry expanse – very little precipitation
- Flora and fauna – very unique
- Some deserts – very hot –
 - Sahara Desert – Africa
 - Thar desert – Northwest India
- Others – cold –
 - Gobi Desert – Asia
 - Antarctica continent – also considered as desert by some
- Harsh living conditions – BUT – humans – adapted to deserts
- India – Thar desert – communities – rich cultural traditions – folk songs, legends – rel. to desert
- Diverse ways – humans – made – all landforms – their home – testimony – adaptability
- *Tinai* – ancient Tamil Sangam poetry – (*tinai* – landscape – main occupation) –
 - *Kurinji* – mountainous regions – hunting and gathering
 - *Mullai* – grasslands and forests – cattle rearing
 - *Marudam* – fertile agriculture plains – farming
 - *Neydal* – coastal regions – fishing and seafaring
 - *Palai* – arid, desert-like regions – journeying and fighting